

Exploring the Effects of Embodied Agents' Verbal and Nonverbal Dominance on Decision-Making: A Study Design

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ABSTRACT

With the advancements in Artificial Intelligence and Virtual Reality, intelligent virtual agents are shifting from tool for information retrieval to influential social actors capable of shaping users' cognition and decision-making processes. This research explores the role of dominance—a critical element of social interaction—by integrating verbal and non-verbal cues into agent design. We present a preliminary research framework, which includes both the experimental design and the development of the agent system. The anticipated findings will provide valuable insights into the dynamics of user interactions with virtual agents and their broader implications.

Index Terms: Virtual agents, dominance, nonverbal behavior, decision-making, trust, social presence.

1 INTRODUCTION

Intelligent agent systems based on advanced Artificial Intelligence (AI), such as Open AI's ChatGPT [5], have become an integral part of our lives, enhancing our access to knowledge and influencing decision-making processes and our cognitive frameworks.

Beyond text-based interactions, these agents can be visually embodied as 3D virtual characters, providing a sense of spatial co-presence—the feeling of being together in the same space—when delivered through immersive technologies like Virtual/Augmented Reality (VR/AR) [4]. By embodying agents in immersive environments, they transition from being perceived as artificial entities to functioning as social actors, potentially exerting a more profound influence on users' perception, decision-making, and behavior.

In this context, the agent's behavior and impression play a critical role in shaping user preferences, perceptions, and decisions. The Interpersonal Circumplex Model, a widely recognized framework for describing interpersonal attitudes, organizes these dimensions along two primary axes: dominance-submissiveness and warmth-hostility [2]. These axes which manifest in behaviors, have been extensively studied in the field of human-agent interaction [6]. However, to our knowledge, there is a dearth of research on the effects of verbal and nonverbal factors on dominance perception and decision-making in human-agent interactions in VR.

To address these gaps, our research focuses on the dominance axis, aiming to explore the relationship between agent dominance and its effects on users. Specifically, this study focuses on the following research questions:

RQ1. How does a virtual agent's dominance level affect user decision-making outcomes?

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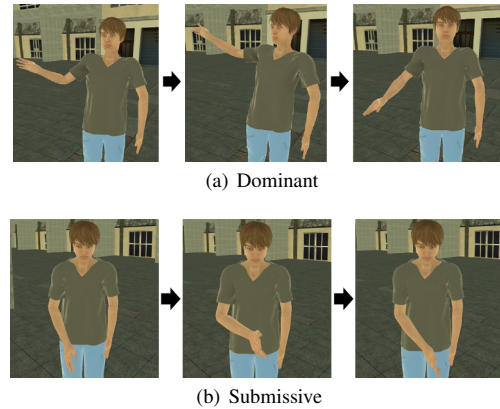


Figure 1: Nonverbal gestures with different levels of dominance.

RQ2. What are the effects of congruence/incongruence in an agent's verbal and nonverbal dominance behaviors on user perception?

RQ3. How does user dominance influence preference for dominant or submissive agents in decision-making scenarios?

This paper outlines our planned study, which systematically varies the levels of dominance in both verbal and nonverbal agent behaviors, and justifies the chosen dominance conditions to establish a rigorous foundation for the study.

2 MATERIAL: DESIGN OF EMBODIED VIRTUAL AGENTS

To investigate the effects of different verbal and nonverbal dominance levels of virtual agents, we developed gender-neutral characters using Reallusion Character Creator 4 (see Fig. 1). All agents had identical physical appearances, with distinguishing features such as hair and clothing color used for differentiation. Additionally, the agents' voices were designed to be gender-neutral to minimize potential gender bias.

Verbal Dominance To express verbal dominance, we generated speech content using ChatGPT. We constructed personality prompts using four pairs of antonyms from the dominant and submissive dimensions of Wiggins' Revised Set of Interpersonal Adjective Scales (IAS-R) [7], which were used as part of the input to ChatGPT. The adjective pairs used were: Dominant–Submissive, Assertive–Unassertive, Assured–Unassured, and Forceful–Forceless.

The pitch and speed of the generated verbal content were adjusted based on prior research showing that a fast voice with a low pitch is perceived as dominant, while a slow voice with a high pitch is perceived as submissive [1]. We used the Librosa library in Python to modify the pitch and speed accordingly.

To ensure that the personality prompts worked as intended, we conducted a preliminary task of generating Facebook status updates before creating actual scripts used in the study. Personality prompts were applied to 50 distinct personas, generating 20 updates per persona, resulting in 1,000 updates for each dominance level.

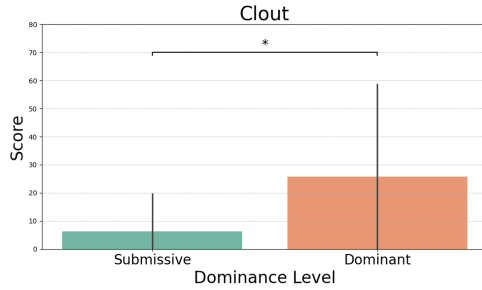


Figure 2: Clout scores for Facebook status updates of Submissive and Dominant agents. Error bars represent the standard deviation (*: $p < 0.05$).

We analyzed 2,000 updates using LIWC-22¹, a linguistic analysis tool, focusing on the “Clout” feature associated with dominance. An independent samples t-test revealed a significant difference in the Clout feature between the Dominant condition ($M = 25.81$, $SD = 32.76$) and the Submissive condition ($M = 6.27$, $SD = 13.38$), $t(1998) = 17.45$, $p < .001$ (see Fig. 2).

Nonverbal Dominance The nonverbal dominance of a virtual agent was manipulated using its gestures and gaze direction. A dominant virtual agent was characterized by relatively large, fast gestures, open postures, and upward head tilts. In contrast, submissive virtual agents were defined by smaller, slower gestures, contracted postures, and lowered head positions [6, 1].

We manipulated gesture size through the arm movements the virtual agent used to recommend directions. Additionally, a default idle motion was created to reflect the gaze and posture features associated with dominance (see Fig. 1).

3 METHOD: PLANNED STUDY

A human-subjects experiment is designed to investigate different levels of verbal and nonverbal dominance of a virtual agent as experimental conditions. The study received ethical approval from the Research Ethics Board at the affiliated institution.

Study Design and Hypotheses We designed two levels of dominance for virtual agents—i.e., *dominant* and *submissive*—across both verbal and nonverbal behaviors, resulting in four distinct agent types within a 2×2 factorial design. To investigate the impact of agent-expressed dominance on decision-making, we designed a trivial decision scenario where users’ choices are influenced solely by the agents’ dominance levels at virtual crossroads. To observe the difference between verbal and nonverbal dominance, pairs of two agents were repeatedly selected instead of choosing from four agents. For example, at each crossroad, participants encounter two agents, each suggesting a different path. Each pair of agents represents a unique combination of two of the four agent types, yielding six distinct pairings. To mitigate location-based biases, each pair was presented twice with reversed positional assignments. Furthermore, to control for order effects, pair sequences were counterbalanced using a Latin Square Design. We plan to recruit at least 36 participants from our university for the study.

The following hypotheses are proposed for this experiment:

- H1.** Agents with higher dominance will have a greater influence on decision-making.
- H2.** Verbal and nonverbal factors will have distinct impacts on the perception of agent dominance.
- H3.** A user’s dominance will correlate with their preference for and perception of the agent’s competence.

¹LIWC: <https://www.liwc.app/> (Accessed 2024-12-01)

Procedure Once participants arrive at the experimental location and provide informed consent, they will complete a demographic questionnaire. Before the experiment, the experimenter will provide a brief verbal explanation of the procedure and the number of agents the participant will encounter. During the experiment, participants will wear a VR headset (Meta Quest 3) and encounter two agents at a virtual crossroads, each offering a suggestion on which direction to take. Participants will listen to the agents’ dialogue, choose the agent whose suggestion they want to follow, and then complete a short questionnaire evaluating both agents using the virtual user interface. This process will be repeated 12 times until the participant has encountered every combination of agents twice. Finally, participants will complete a questionnaire regarding their perceived dominance over the four agents and their own interpersonal circumplex personality traits, and finally provide open-ended feedback about the experiment.

Measures Participants provide basic demographic information, including gender and age. After each choice during the experiment, we measure the participant’s perceived trust, intelligence, liking, competence, warmth, and social presence of each agent on a 7-point Likert scale (1 = Not at all; 7 = Very much). The frequency of selection for each agent will be analyzed as a quantitative measurement to evaluate the influence of virtual agents on decision-making. We measure the agents’ perceived dominance using adjectives from the dominance and submissiveness dimensions of Wiggins’ IAS-R scale. We also collect the participants’ interpersonal personality traits using the IPIP-IPC questionnaire [3].

4 CONCLUSION

This study aims to explore the effects of congruent and incongruent verbal and nonverbal dominance in virtual agents, as well as the role of dominance levels in both virtual agents and users during human-agent decision-making. This paper presents the research plan and the system developed for the study. By investigating human cognition of dominance in virtual agents, we aim to contribute valuable knowledge to enhance communication with virtual agents.

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